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DUNMAN HIGH SCHOOL **Preliminary Examination** **Year 6**

H2 BIOLOGY

9648/03

Paper 3 Applications Paper and SPA Planning Task

23 September 2013

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A:

Consists of 4 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 1 SPA Planning Task

Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Section C:

Consists of 1 Free-Response Question.

Write your answers on the separate **writing papers** provided. At the end of the examination, fasten all your work securely together.

Sections A, B and C are to be submitted **separately**.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

For Examiner's Use	
Section A [40]	
1	/ 12
2	/ 8
3	/ 13
4	/ 7
Section B [12]	
Section C [20]	
Total [72]	

This document consists of **13** printed pages and **1** blank page.

Section A: Structured Questions (40 marks)

Answer **all** questions in this section.

For
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use

Question 1

Fig 1.1 shows two plasmids pAMP and pKAN. Amp^r and Kan^r confer resistance to the antibiotics ampicillin and kanamycin respectively.

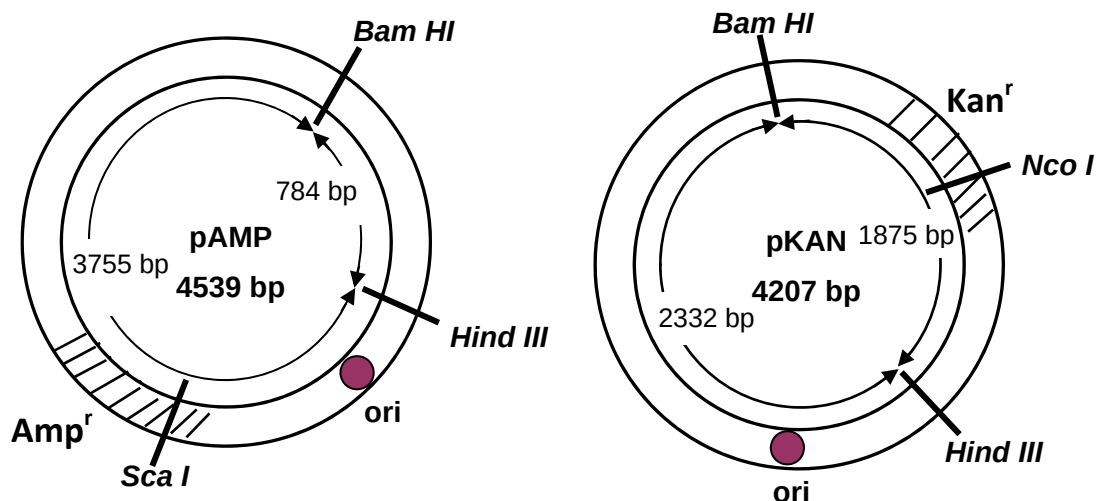


Fig 1.1

In an experiment to make a cloning vector, solution containing the plasmids pAMP and pKAN were treated with a solution containing restriction enzymes BamHI and HindIII. DNA ligase was then added and the mixture was incubated at 37°C.

This experiment produced the plasmid pAK which can be used as a cloning vector. The restriction map of pAK is shown in **Fig 1.2** and the specific recognition sequence of its restriction sites are shown in **Table 1.3**.

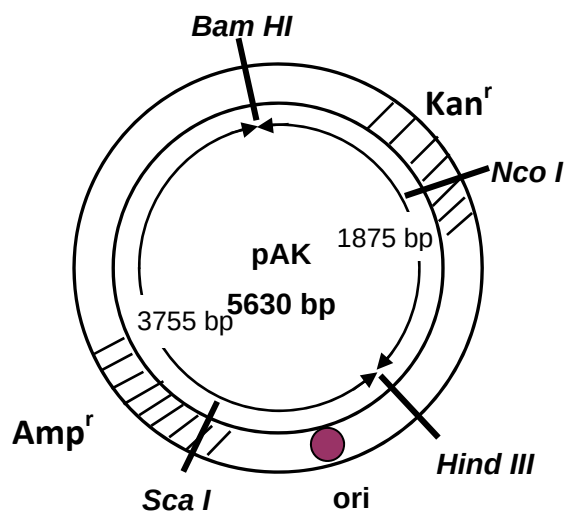


Fig 1.2

Restriction Enzyme	Recognition Sequence
Bam HI	5'-G [^] AG ATC C-3' 3'-C CTA G [^] G-5'
Hind III	5'-A [^] AA GCT T-3' 3'-T TCG A [^] A -5'
Sca I	5'-A G T [^] AC T-3' 3'-T C A [^] TG A-5'
Nco I	5'-C [^] AC ATG G-3' 3'-G GTA C [^] C-5'

^ indicates where the restriction enzyme cuts

Table 1.3

(a) Restriction sites are palindromic. Explain the meaning of the term “palindromic”. [1]

(b) With reference to **Fig 1.2** and **Table 1.3**, explain why *Nco* I is a more suitable choice of restriction enzyme for use in genetic engineering with the pAK plasmid than *Sca* I. [2]

To clone a human gene with a molecular size of 600 bp in bacterial cells, the following procedure was carried out as shown in **Fig 1.4**.

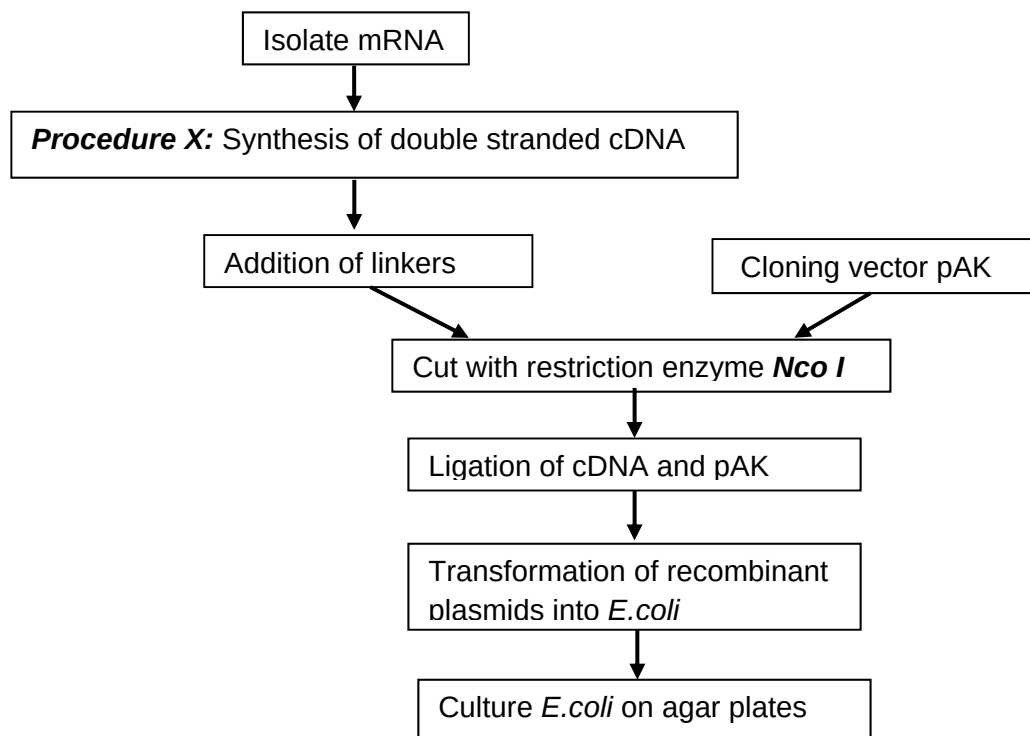


Fig 1.4

(c) With reference to **Fig 1.4**, briefly describe procedure X. [3]

(d) Draw **three** possible gene products formed in the ligation mixture. [3]

- (e) Explain how the plasmid pAK can aid in the successful selection of recombinant colonies with the gene of interest. [3]

Total: [12]

Question 2

For
Examiner's
use

A length of DNA from one of a pair of homologous chromosomes is shown in **Fig 2.1**. The target sites of *EcoRI* are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

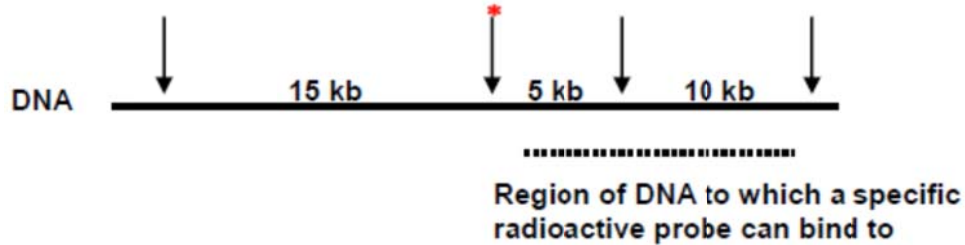


Fig 2.1

A mutation alters one base of the coding sequence at the site marked with an asterisk (*). DNA from two individuals, one who is normal and the other who is heterozygous for this mutation is cut with *EcoRI* and the DNA fragments separated by gel electrophoresis.

(a) On **Fig. 2.2** below,

- (i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individuals tested and label the size of the fragments produced. [2]
- (ii) Circle the fragments which will combine with the radioactive probe and appear as black bands on an autoradiography. [1]

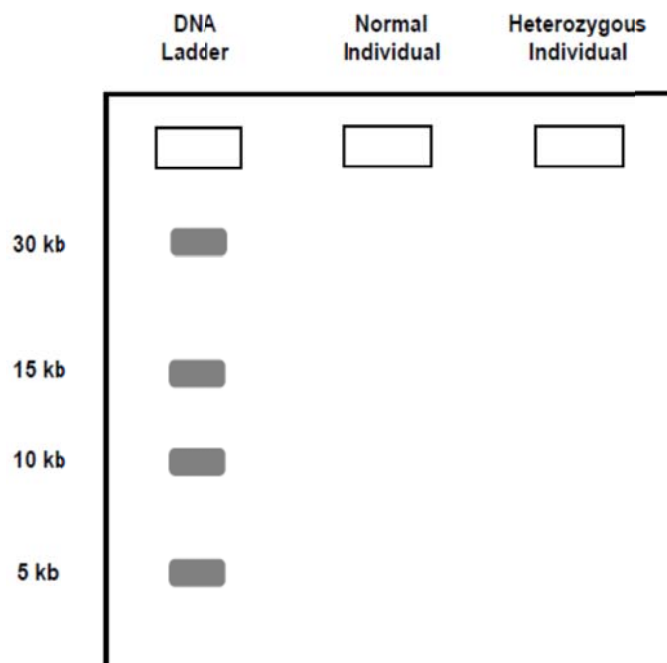


Fig 2.2

(b) Describe how DNA fragments may be separated by electrophoresis. [4]

(c) State the role of the DNA ladder seen in **Fig 2.2**. [1]

Total: [8]

Question 3

For
Examiner's
use

- (a) Cystic fibrosis (CF) is a genetic disease which affects many parts of the body. This disease is caused by mutations of a gene coding for a transmembrane protein (CFTR) which acts as an ion pore.

A study on ion transport was carried out on pancreatic cells by comparing ion transport in cells expressing the normal allele for CFTR with that in cells expressing mutant alleles. The pancreas secretes pancreatic juice which is alkaline to neutralize the acidic gastric acid from the stomach.

The abilities of the cells to transport HCO_3^- and Cl^- were measured and expressed as a $\text{HCO}_3^- : \text{Cl}^-$ transport ratio. The results are shown in **Table 3.1**.

CF allele expressed by cells	$\text{HCO}_3^- : \text{Cl}^-$ transport ratio of cells
Normal alleles	1.0 : 1.0
Mutant alleles associated with CF with inadequate functioning of the pancreas	All < 0.1 : 1.0

Table 3.1

- (i) With reference to **Table 3.1**, explain the cause of inadequate functioning of the pancreas in CF. [2]
-
-
-
- (ii) Suggest **one** possible consequence of inadequate functioning of the pancreas as a result of CF. [1]
-
-
- (b) Gene therapy involves transferring normal CFTR genes into lung epithelial cells with defective genes. Describe how a liposome can be used to insert the normal CFTR gene into a CF patient. [2]
-
-
-

- (c) The mucus produced in the lungs of someone with cystic fibrosis contains a lot of DNA from dead cells. DNase, an enzyme which cuts DNA into short pieces has recently been approved for the treatment of cystic fibrosis. In an investigation, different concentrations of DNase were added to mucus collected from people with cystic fibrosis. **Fig. 3.1** shows the results.

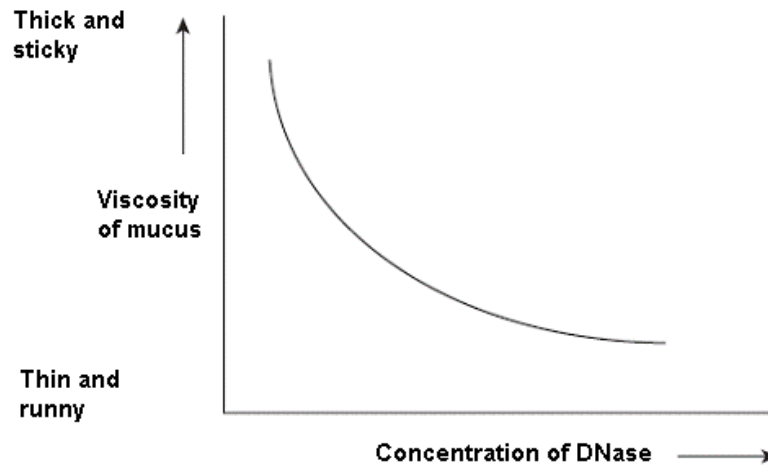


Fig. 3.1

Using the information given in **Fig. 3.1**, explain why inhaling DNase helps a cystic fibrosis patient to breathe more easily. [3]

- (d) Describe how the isolated CFTR gene can be obtained from the genome using the polymerase chain reaction. [3]

- (e) About twenty percent of the DNA molecules produced by the PCR are copied inaccurately. Explain why it is not safe to use the PCR to clone the normal CFTR gene for use in treating cystic fibrosis. [2]

Total: [13]

Question 4

For
Examiner's
use

EH1 is a hybrid derived from a cross between *Eucalyptus grandis* ($2n = 22$) and *Eucalyptus urophylla* ($2n = 22$). Embryo culture is used to develop **EH1** embryos into plantlets. At three different stages of the embryonic development as shown in **Fig. 4.1**, the nutrients and plant growth hormones concentrations in the medium are changed to allow development into plantlets.

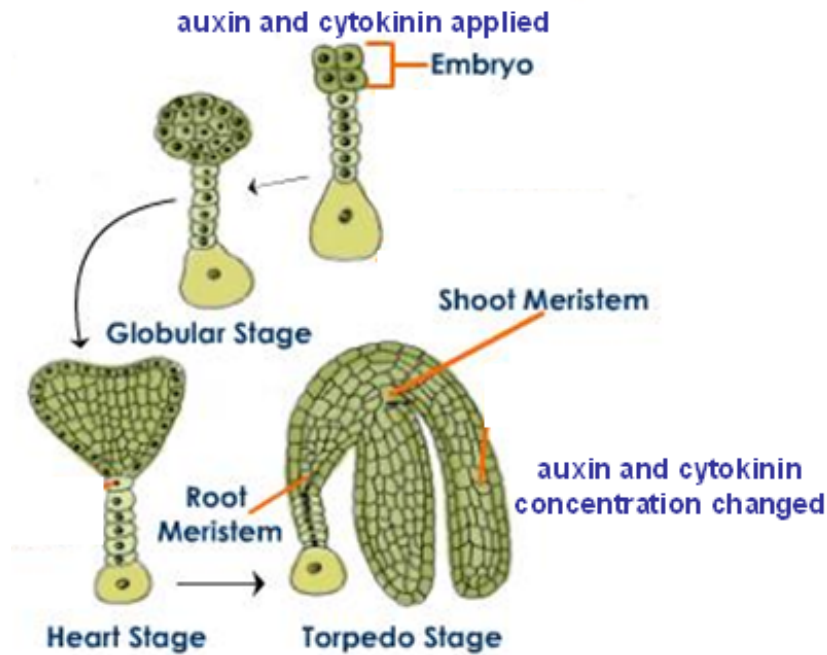


Fig. 4.1

- (a) (i) Explain why the **EH1** plantlets grown from different hybrid embryos are not genetically identical. [2]

- (ii) With reference to **Fig. 4.1**, explain why the concentrations of the plant growth hormones have to be applied and changed to ensure development of plantlets. [3]

The average growth of wood was recorded for EH1 grown in different parts of the United States. Non-GM EH1 was grown in plot A, while GM EH1 was grown in plots B, C, and D. The results are shown in **Fig. 4.2**.

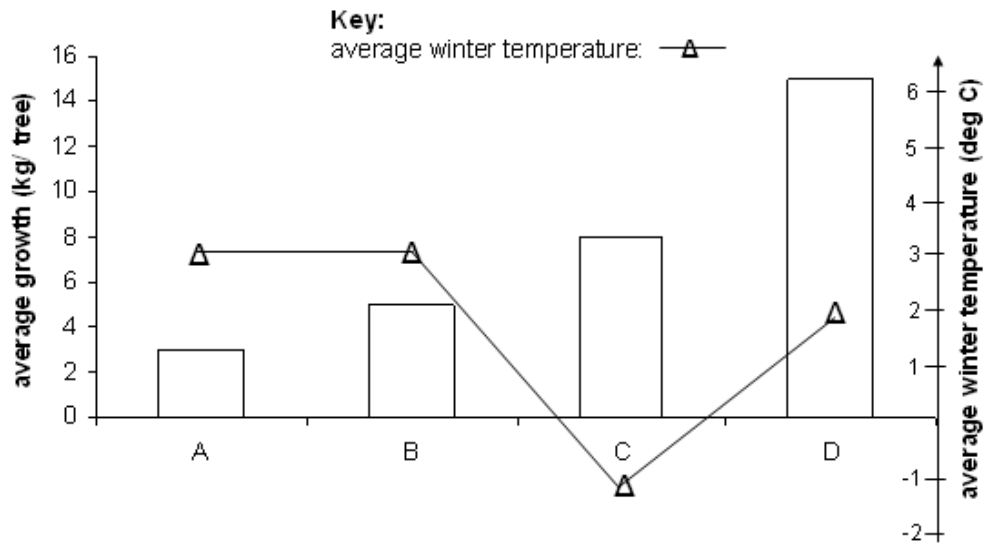


Fig. 4.2

- (b) Explain how the data shows that lignin gene expression level is difficult to control in GM crops despite being placed under the control of an antifreeze promoter. [2]

Total: [7]

Section B: SPA Planning Task (12 marks)

Write your answers on the writing paper provided.

A **NIL RETURN** is required.

Question 5

Digestion of lipids usually occurs in the small intestines by the action of pancreatic lipase. Unfortunately, dietary fats do not dissolve in water, and bile salts secreted by the liver are required to emulsify fats. This provides a larger surface area for lipase to act on.

Phenolphthalein is an acid-base indicator which is colourless in acidic solutions and pink in basic solutions such as sodium carbonate solution. A 1:1 ratio of sodium carbonate solution to full cream milk will provide a slightly alkaline pH which is optimum for lipase action.

Using this information and your own knowledge, design an experiment to investigate the effect of bile salt concentration on the rate of digestion of lipids.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Phenolphthalein
- 5% lipase solution
- 5% Bile salts solution
- Sodium carbonate solution
- Distilled water
- Full cream milk
- Stopwatch
- A variety of different sized beakers, measuring cylinders and syringes for measuring volumes
- Access to standard laboratory equipment

Your plan should have a clear and helpful structure to include:

- An explanation of theory to support your practical procedure
- A description of the method used, including the scientific reasoning behind the method
- An explanation of dependent and independent variables involved
- How you will record your results and ensure they are accurate and reliable as possible
- Proposed layout of results tables and graphs with clear headings and labels
- The correct use of technical and scientific terms

Total: [12]

Section C: Free-Response Questions (20 marks)

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 6

(a) Describe the treatment of SCID using gene therapy. [10]

(b) Discuss the social and ethical considerations for gene therapy. [10]

Total: [20]

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**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2013
YEAR SIX
H2 BIOLOGY (9648)
PAPER 3**

Answer Scheme

Structured Answers

1(a)

nucleotide sequences that are the same on both strands of the DNA molecule when read in the same direction (5' to 3') / read the same in the 5' to 3' direction on each strand of DNA.

(b)

Nco I produces staggered / sticky ends whereas Sca I produces blunt ends;

This facilitates the annealing of the gene of interest to plasmid when the plasmid and gene are cut with Nco I by complementary base pairing;

Using Sca I would require the additional procedure of adding linker DNA to generate sticky ends to facilitate annealing of gene of interest.

(c)

Oligo-dT primers anneal to 3' poly-A tail of mRNA

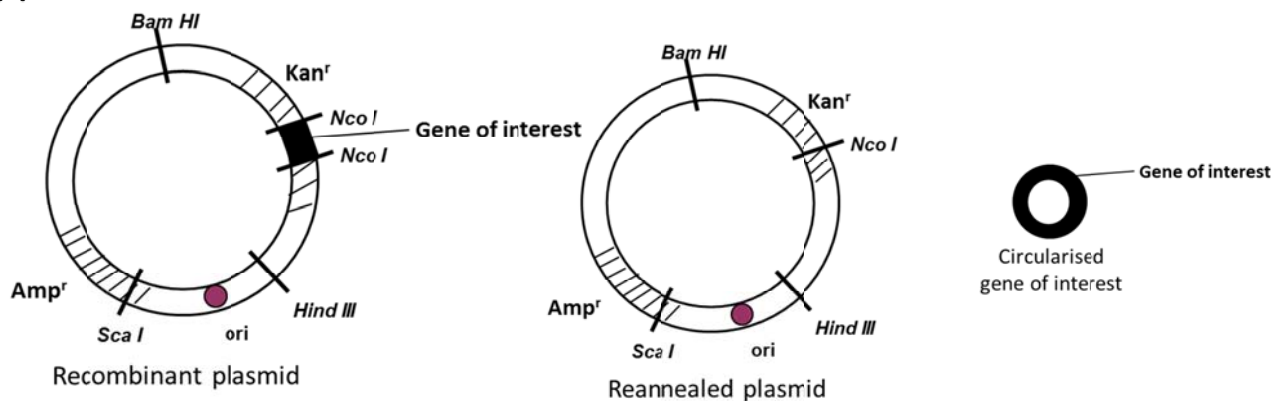
Reverse transcriptase extends primers using mRNA as template to synthesize single stranded cDNA, product is RNA-DNA duplex.

Partial digestion of mRNA using alkaline solution / RNase H

ref to synthesis of 2nd complementary strand to form double stranded cDNA using DNA polymerase

3 max.

(d)



1 mark each

(e)

Amp^R gene codes for ampicillin resistance thus allows for first selection for successfully transformed colonies which have taken in the plasmid;

Colonies with reannealed plasmids will have intact Kan^r gene which codes for kanamycin resistance and will survive on kanamycin plate;

Colonies with recombinant plasmids will gene of interest inserted into Kan^r gene which results in insertional inactivation, and will not survive on kanamycin plate;

2(a)

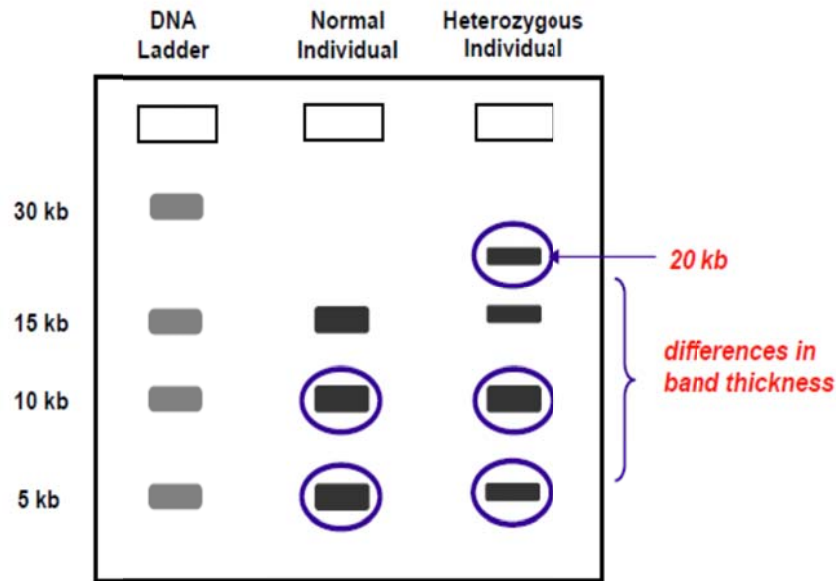


Figure 2.2

(i)

Normal individual – 3 correct bands, labeled and of the correct thickness;

Heterozygous individual – 4 correct bands, labeled and of the correct thickness;

(ii)

circling of 10kb and 5kb bands for normal individual & circling of 20kb, 10kb and 5 kb bands for heterozygous individual;

(b)

A loading dye is added to the DNA samples to help DNA to sink into the wells / allows the process of electrophoresis to be tracked;

DNA ladder and samples are loaded into wells;

Gel is submerged in a buffer solution that will conduct electricity;

electric current is applied though electrodes at opposite ends of the gel;

negatively-charged DNA fragments migrate to the positively charged electrode / anode;

Gel electrophoresis separates DNA fragments by size / molecular weights, with the smallest fragments moving the fastest and furthest;

4 max.

(c)

As a reference / standard, providing known DNA fragments of specific lengths to estimate the size of DNA molecules in samples

3 (a) (i)

Transport ratio is all $< 0.1 : 1.0$;

Inability to transport HCO_3^- out of pancreatic cell;

increase in acidity / decrease in pH outside the cells in the intestine;

2 max

(ii)

Poor digestion of proteins/ lipid/starch in the intestines thus poor absorption of nutrients;

Decreases secretion of insulin/glucagon thus affect blood glucose level;

ref. damage to wall of intestine/ pain in the intestine;

1 max

(b)

Normal CFTR gene is inserted into liposomes which are inhaled/sprayed into nose and mouth;

Liposomes fuse with the cell membrane and release normal CFTR gene into (cytoplasm of lung epithelial) cells;

(c)

A higher conc. of DNase will result in a greater number of shorter DNA pieces;

With shorter DNA pieces in the lungs, viscosity of the mucus is reduced;

more air can pass through the airways/airways are no longer blocked/mucus more easily removed from airways;

(d)

Heat dsDNA to 95°C for 1-2min to break the hydrogen bonds (between complementary bases) / separate DNA strands;

Lower temperature to 55°C for 1-2min to allow annealing of forward and reverse primers to ssDNA / to regions flanking the CFTR gene;

Increase temperature to 70°C for 1-2min to allow Taq polymerase catalyses formation of new DNA strand (using the template);

(e)

percentage risk is too high for human application/ ref to a wrong/mutant gene introduced into CF patient;

incorrect mRNA formed changes sequence of amino acids in protein;

produce a toxic/harmful/non-functional/ mutated protein;

chloride ions not transported / thick mucus results;

2 max

4 (a) (i)

Crossing over and independent assortment and random segregation of chromosomes during meiosis;

Formed gametes which showed variation / allelic recombination;

(ii)

Need to apply auxin and cytokinin to stimulate mitosis of undifferentiated embryonic cells;

Need to change to low auxin to cytokinin ratio to stimulate formation of shoot meristem;

Need to change to high auxin to cytokine ratio to stimulate formation of root meristem;

(b)

B, C and D experienced average winter temperature of 3°C , -1°C and 2°C respectively, yet average growth are 5, 8 and 15 kg/ tree respectively;

There is no correlation between temperature and average growth;

Planning Answer

1. theory and expected trend
2. theory to support method to measure rate
3. independent variable
4. dependent variable
5. AVP – temperature with method to control
6. dilution
7. appropriate volumes suggested
8. Measuring time taken for indicator to decolourise
9. 3 replicates, 2 repeats
10. method to calculate rate $1/t$
11. appropriate table
12. appropriate graph
13. correct use of technical and scientific terms

Essay Answers

6 (a) Describe the treatment of SCID using gene therapy.

1. remove T-lymphocytes from patient with SCID and insert healthy genes in them using a **retrovirus**, then transplanting them back into the patient;
2. gene therapy can be used to treat both ADA-SCID and sex-link-SCID;
3. An ex vivo gene therapy approach was used;
4. where lymphocytes obtained from the blood of the patient/bone marrow stem cell;
5. was stimulated to proliferate in culture;
6. A functional adenosine deaminase (ADA) cDNA/ IL2-RG cDNA in a retroviral vector was then introduced into these cells;
7. The retrovirus had been inactivated to become non-pathogenic;
8. The transgenic lymphocytes were allowed to proliferate further;
9. retroviruses naturally integrate their genome into the genome of host cells;
10. thus the retrovirus was integrating the normal ADA gene / IL2-RG gene into the chromosomes of the patient's cells during this culturing;
11. After a short period of culturing, these ADA-containing lymphocytes were re-introduced into the patient;

6 (b) Discuss the social and ethical considerations for gene therapy.

Arguments in favour of gene therapy:

1. Potential for treating ill patients or preventing onset of severe illness
2. corrects the genetic defect present in the individual alone
3. like any other new medical technology.
4. To prevent research on gene therapy is to infringe upon intellectual freedom of researchers.
5. With current rate of progress in gene therapy techniques risks become more identifiable and outcomes can be predicted rather precisely.
6. offers true cure, not just a relief of symptoms or a palliative treatment.
7. The expense of somatic-cell therapy for multiple generations is avoided.
8. Medicine should respond to the reproductive health needs of prospective parents at risk for transmitting serious genetic diseases.
9. The scientific community has a right to free inquiry within the bounds of acceptable human research.
10. Once successful techniques are developed, gene therapy could help parents and researchers avoid the moral dilemma of disposing "defective" embryos in the laboratory

Arguments against gene therapy:

11. It is difficult to distinguish between "good" and "bad" uses of gene modification techniques.
12. It is important to consider between genetic disease and genetic attribute. Consideration should be given beyond the therapeutic if there is only a genetic tendency towards a particular disease.
13. There is potential for harmful abuse of the technology.
14. There are many unresolved questions raised by gene therapy:
15. There is difficulty in following patients in long-term clinical research, including surveillance of future generations to monitor the long-term effects of gene therapy.
16. Many gene therapy candidates are children who are too young to understand ramifications of gene therapy
17. There are potential conflicts of interest of individuals who wish to protect their privacy and confidentiality versus interests of insurance companies or society in not bearing the financial burdens of caring for children with serious genetic defects.
18. There are issues of justice and allocation: can we afford the expense of gene therapy? Who should receive gene therapy? If only those who can afford it, where is the principle of justice? The procedures now being tested are expensive and require expertise and equipment found only in major medical centers. A related question is, "Should gene therapy be reserved for treating serious diseases?" "Who should pay for their use?" "Should we skew the distribution of desirable biological traits?"
19. Long-term effects of germ-line therapy are not known.
20. It opens the door to attempts at altering human traits not associated with disease (i.e., eugenics), which in turn leads to issues of increased social inequality and a lowered tolerance for genetic diversity.
21. Violates rights of future generations to inherit a genetic endowment which has not been intentionally modified.
22. Too expensive.
23. Essentially mates generations of unconsenting research subjects (i.e., germ-line therapy involves research on early embryos and affects their offspring).